



Agent Foundations

Daniel C

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Why Agent Foundations?

Aligning a system that doesn't exist yet

- Alignment may eventually require us to steer a system far more capable than anything we have built, under conditions unlike anything we have yet observed
- Agent foundations studies the properties that capable agents tend to share *in general*, rather than the details of any one system – giving us a way to reason about highly capable agents *in advance* of being able to study them directly
- At a sufficiently dangerous capability level, we may need to get key safety properties right on the *first critical try*: a serious failure could be unrecoverable, leaving no room for the trial-and-error that science normally relies on
- Even if we learn about capable agents empirically from weaker systems, we still need a theory of *which* properties continue to hold as capability increases – and supplying that theory is itself the work of agent foundations



Reflective Stability

Self-modification and the stability of goals

- A sufficiently capable agent may be able to: rewrite its own source code, build a more capable successor, and radically revise its world model (e.g. learning new physics, ontology shifts)
- A property is *reflectively stable* if it remains invariant under all such self-modifications
- A safety property we instil only matters if it is reflectively stable: a guardrail is of little use if the agent can self-modify to remove it, or build a successor that lacks it
- For a sufficiently capable goal-directed agent, the working assumption is that **a property persists only if there is some reason it has to** – architectures, constraints, and conceptual schemes are all candidates for being optimised away under enough pressure
- So to guarantee a safety property holds (e.g. “this system does not cause a catastrophe”), it is not enough to instil it once; we need it to be *invariant under self-modification*, and we need to understand what kinds of properties *can* have that invariance



Two Pathways of Impact: Modelling vs. Implementation

Modelling

- Build an *abstract mathematical model* of an idealised, highly capable agent, and use it to make safety claims about how such an agent would behave
- The model need not be runnable; its value is that it lets us reason about such agents *in advance*, and in particular show *why* a given alignment proposal would fail

Implementation

- Develop a theory to the point where it could be turned into an actual algorithm or running system that we could build and inspect
- On this pathway, researchers often favour *modular* architectures – a separate, inspectable world model, a planning module, and an explicit representation of the goal – rather than one opaque network, so that each part can be checked and controlled on its own
- A clean factorisation like this is what would let us point at “the goal” or “the world model” inside the system and verify or edit it directly

(Distinction following C. Wyeth, *Modeling versus Implementation*.)



Reasoning About Ideal Intelligence

Coherence theorems – and a too-cheap objection

- Coherence theorems (money-pump / Dutch-book arguments) give *some* justification for modelling a capable agent as an **expected-utility maximiser**: an agent with exploitable, incoherent preferences leaves value on the table, so optimisation pressure pushes towards coherence
- But as stated this is **almost vacuous**: *anything* can be cast as a utility maximiser. A rock is an expected-utility maximiser whose utility is maximised by “sitting on the floor”
- A frame that fits *every* system equally makes *no* predictions. To get predictive content we must say what the agent is coherent **over**



Reasoning About Ideal Intelligence

Coherence over resources makes concrete predictions

- Utility maximisation becomes *predictive* once we claim a physical system is well modelled as a maximiser over **specific resources** – money, energy, free energy, . . .
- Define preference operationally: the system **prefers** state A to state B if it *expends resources* to move from B to A
- Coherence then predicts something **falsifiable**: an *efficient* agent will not spend resources to go $A \rightarrow B$ and also spend resources to go $B \rightarrow A$ – it does not “throw away resources” by pumping itself around a cycle
- And this is **checkable**: for a given system and resource, just look. Does a cell spend energy converting compound $X \rightarrow Y$ and energy converting $Y \rightarrow X$? Coherence theorems say it is useful to model a system as an efficient agent exactly when it does not waste resources like this



Why care about resources?

Instrumental convergence

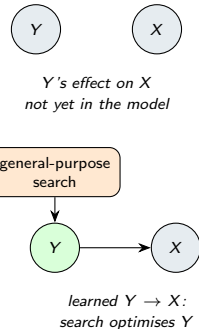
- Why single out “resources” at all? **Instrumental convergence**: for a *wide variety* of terminal goals, a wide variety of resources are useful for achieving them
- So, all else equal, **accumulating resources** is “good” by the agent’s own lights – no matter what its terminal goal is. Resources are the goal-independent currency of getting things done
- Physics echoes this: **resource theory** formalises a resource as a *monotone* – a state function (e.g. free energy, negentropy) that cannot increase under the cheap/“free” operations
- Having more of the resource buys more **optionality**: a wider set of future states you can still steer towards. More resources = more control over the future



General-purpose search

Why intelligence is “general purpose”

- Why expect a capable mind to be *general-purpose* – able to steer towards a *wide variety* of goals, not just one fixed task?
- Suppose you hold a terminal goal X , but your world model is **incomplete**: some variable Y causally influences X , yet you are not (yet) aware of that pathway
- As you learn and update your world model, you discover the path $Y \rightarrow X$ – i.e. Y is revealed to be an **instrumental subgoal** for X . Many variables may “turn out” this way
- Now suppose the agent runs a **general-purpose search**: an internal algorithm that (1) takes *any* goal description and (2) returns a plan to achieve it. Then the moment a new instrumental subgoal is discovered, the agent can optimise for it **on the fly** – it need not have anticipated it

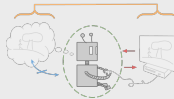




Dualistic agents

The cybernetic picture: Alexei plays a video game

- In the standard (“cybernetic”) picture, an agent learns by building a *model* of its environment, and uses that model to plan towards a goal. This is Alexei, playing a video game:



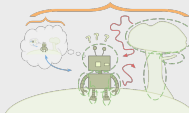
- **Clear input & output channels** between the agent and the environment
- The agent is **“larger” than the environment**: it can hold a full copy of the game inside its own mind
- The agent is **“outside” the environment**: it need not reason about itself, only about optimising the environment
- Standard idealised models of agency – an expected-utility maximiser, or any agent that keeps a full internal model of the world and plans within it – assume this dualistic, “large agent” picture
- But can we still reason about an agent that is *smaller* than its environment, and contained inside it?



Embedded agents

Embedded agents: Emmy plays real life

- This is Emmy. Emmy is playing real life:



- The environment **contains her**: Emmy is just one part of the environment, made of the same pieces as everything else
- **No well-defined input/output channel** cleanly separating agent from environment
- She **cannot hold the environment inside her own head** – the environment is bigger than she is



Embedded agents

Why embeddedness is hard

- The environment may contain **copies of her**, and other equally capable agents that model her while she models them
- She may be **modified, destroyed, or copied** (by herself or by others) between interaction steps
- Being inside the environment she manipulates, she is also capable of **self-improvement**
- Dualistic models struggle to represent this: a model that treats the environment as simpler than the agent itself cannot contain a copy of the agent, so it cannot describe an agent that is part of its own environment
- We would like a theory of consequentialist agency that is consistent with this embedded setting



Self-Modification and Vingean Reflection

Studying self-modification in a simplified setting

- We want to study **self-modification**: a sufficiently capable agent A_1 may rewrite its own code or build a successor A_0 more capable than itself (self-improvement is a special case of building a successor)
- Strategy: reduce this to a simpler setting we can analyse cleanly, but whose lessons should carry over to richer settings
- **Vingean uncertainty**. Here “more capable” means A_0 makes more accurate predictions, or selects more effective actions towards a goal, than A_1 . So A_1 *cannot* predict exactly what A_0 will do – if it could, it could simply take those actions itself and be equally effective, and A_0 would not be more capable after all (cf. Deep Blue’s designers knowing it was “trying to win” without knowing its moves)
- **Implication**. A_1 must establish, in an *abstract* sense, that building A_0 is safe – without simulating A_0 step by step

The tiling principle

- We want A_1 to be able to approve building a successor *similar to itself* – e.g. the same algorithm running on different hardware
- This substrate-independence is the point: we want a *robust* abstract argument (“constructing this successor is safe”), not a fragile rule that only blesses agents with identical source code



Self-Modification: Why a Logical Framework?

Stripping away empirical uncertainty

- A real-world agent also has **empirical uncertainty**: which environment am I in? how will my actions affect it?
- To isolate the self-trust problem, we *strip this away*: assume the agent already knows its environment and how its actions influence it
- For simplicity, assume both the agent and the environment can be represented as **programs**

Why proofs about programs are the right tool

- **Concrete behaviour becomes provable.** If a program halts and returns an output, L can prove it – the proof just lists the execution trace
- **But proofs also certify general behaviour.** One can prove a program never halts, or prove a universally quantified claim (“for all inputs, the output satisfies X ”) – things no single finite trace can show
- So once agents and environments are programs, proofs are *strictly more general* than simulation: they let us certify a property *in the abstract*, without inspecting the execution step by step – exactly what Vingean uncertainty demands



How can A_1 trust A_0 ?

From “ A_0 acts on its proofs” to “ A_0 ’s actions are safe”

- A_1 and A_0 share a satisficing goal $\mathcal{G}$, and A_0 only takes actions it can *prove* achieve $\mathcal{G}$
- A_1 inspects A_0 ’s source code and verifies: if A_0 is constructed, then any action A_0 takes was proved (in A_0 ’s proof system) to achieve $\mathcal{G}$:

$$A^1 \Vdash \bar{A}^0 \rightarrow (\forall b_i \in \text{Acts}^0 : \bar{b}_i \rightarrow \Box_0[\bar{b}_i \rightarrow \mathcal{G}]) \quad (2.3)$$

($\bar{A}^0$ = “ A_0 is constructed”, $\Box_0[\phi]$ = “a proof of ϕ exists in A_0 ’s system”)

- But this only tells A_1 that A_0 *acts on what its proof system certifies* – not that what it certifies is actually *true*
- To close this gap, A_1 needs to trust A_0 ’s proof system: whenever A_0 proves something, it is actually true

$$A^1 \Vdash \forall b_i : (\Box_0[\bar{b}_i \rightarrow \mathcal{G}] \rightarrow (\bar{b}_i \rightarrow \mathcal{G})) \quad (2.4)$$

- Combining (2.3) and (2.4): $A^1 \Vdash \bar{A}^0 \rightarrow (\forall b_i \in \text{Acts}^0 : (\bar{b}_i \rightarrow \mathcal{G}))$
 “If A_0 is constructed, then any action A_0 takes achieves the goal”



The Löbian Obstacle

The problem of self-trust

- When A_1 and A_0 use the *same* proof system L , (2.4) becomes $L \vdash \forall \phi : (\Box \phi \rightarrow \phi)$: “whatever I can prove is true” – L vouching for its own proofs
- This conflates two distinct things. $L \vdash P$ means a concrete proof of P is in hand. $\Box P$ is the *internal* statement “a proof of P exists”, which L can establish abstractly (e.g. by reasoning about a proof-search program) *without exhibiting* any such proof
- **Löb’s theorem.** For any statement C : if $L \vdash \Box C \rightarrow C$, then $L \vdash C$. The only statements L can vouch for in the abstract are ones it can already prove outright
- E.g. self-trust would give $L \vdash \Box(2+2=5) \rightarrow (2+2=5)$, and Löb’s theorem then forces $L \vdash 2+2=5$ – impossible for a consistent L
- So a consistent system cannot trust its own proofs *in the abstract*; it can only trust a proof once it has *witnessed it directly*



The Löbian Obstacle

Consequence for tiling agents

- One workaround: A_1 can trust A_0 's proofs if A_0 uses a *strictly weaker* proof system than A_1 (a stronger system can trust a weaker one)
- But then each successor in a chain must use a weaker system than its predecessor – a “telomere” of logical strength that shortens with each generation
- Eventually the chain runs out of trust, so indefinitely long chains of safe self-modification seem blocked. The *tiling agents* programme studies how, and whether, this obstacle can be overcome



Embedded Subproblem: Logical Uncertainty

Empirical vs. logical uncertainty

- **Empirical uncertainty:** “will it rain tomorrow?” – we lack *evidence* about the world; more observations would help
- **Logical uncertainty:** “is the millionth digit of π a 7?” – the answer is already fixed by the axioms of mathematics. We are not missing observations; we are missing *computation*
- Other examples: the output of a program whose source code you already have, or whether a specific large number is prime

Why this matters for embedded agents

- Standard Bayesian reasoning assumes *logical omniscience*: the agent instantly knows all consequences of its beliefs, at no computational cost
- A bounded, embedded agent cannot: it may know every axiom of arithmetic yet still be uncertain about what they imply
- We need a principled way to assign probabilities to *logical* facts, and to refine them as we spend more computation



A Theory of Logical Uncertainty

Analogy with Bayesian probability

- Bayesian probability gives us a *model* of how an idealised agent should hold beliefs and make predictions under empirical uncertainty – even though no real agent ever performs exact Bayesian inference
- We would like the same for *logical* uncertainty: a clean account of how a bounded agent should assign and update probabilities over logical facts

Modelling, then implementation

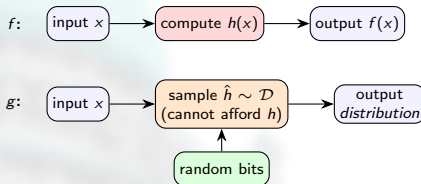
- Recall the modelling-vs-implementation split: a good *theory* of logical uncertainty (the model) can then guide *algorithms* that approximate it
- This mirrors how the theory of Bayesian inference guides practical, approximate inference algorithms (variational methods, MCMC, . . .)



A Model of Logical (Computational) Uncertainty

Ground truth vs. a time-bounded approximator

- **Ground truth:** a computable function f from inputs to outputs (or output distributions)
- **Approximator:** a program g that receives the input *and* a string of random bits, and must answer within a fixed **time bound**. The random bits induce a distribution over g 's outputs; its quality is how close that distribution is to f
- **The tradeoff.** If computing f needs a subroutine h that g cannot afford in time, a sensible move is to use the random bits to *sample* a value $\hat{h}$ from a distribution over h 's output, and propagate that uncertainty forward
- As the time bound grows, g can just compute f directly and the random bits become irrelevant – *uncertainty shrinks as we spend computation*





Logical Uncertainty: Open Problems (I)

An algorithm approximating itself

- Give one algorithm A several possible time budgets. With a short budget, A can try to approximate “*what would I output if I had a larger budget?*”, using random bits in place of the computation it cannot afford
- So the same machinery models an agent reasoning about its own future, more-computed conclusions

Randomness as logical counterfactuals

- In the subroutine picture, the random bits build a distribution over a subroutine's output: each draw is a different *possibility* for what that output “could” be
- We need these possibilities precisely because we lack the compute to pin down the actual value – so different settings of the random bits behave like different **logical counterfactuals**



Logical Uncertainty: Open Problems (II)

Which representations are useful?

- Suppose the approximator may also *condition* on some side information about the input. Useful side information lets it reduce its logical uncertainty
- Handing it the output of an expensive subroutine saves that computation, freeing compute to reduce uncertainty elsewhere
- Handing it an *encryption* of the input – carrying the same Shannon / algorithmic information – helps not at all
- So this gives a handle on which *representations* of the same information are actually useful to a bounded reasoner, and on when “post-processing” a representation is worthwhile



Aside: Logical Induction

One concrete realisation of these ideas

- **Logical induction** (Garraabrant et al.) gives a computable way to assign probabilities to logical sentences that improve as more is proved. Picture a *market* pricing each sentence φ in $[0, 1]$ (its probability); a φ -share pays \$1 once φ is proved
- **The logical induction criterion:** no “efficient” (polynomial-time) trader can make unbounded profit against the market
- (This weakens the classical Dutch-book argument – “no trader can exploit you $\Rightarrow$ your beliefs obey the probability axioms” – to “no *efficient* trader can exploit you”; that single condition already implies many good properties.)
- In the limit it yields: **coherence** (probability 1 to theorems, 0 to contradictions, monotone under implication); **statistical patterns** (it prices “the n th digit of π is 7” near 10% without computing it); and **self-trust** (current credence = weighted average of its expected future credences)



Descriptive Agent Foundations

From normative to descriptive

- **Normative (ideal) agent foundations:** characterizes how a perfectly rational agent *should* reason and act in principle – starts from desiderata of ideal agents and derives what follows
- **Descriptive agent foundations:** asks what agents we actually encounter in the wild actually *look like* – given any physical system (a neural network, a bacterium, a future AI), can we identify its goals, world-model, and decision-making structure?
- The aim is to develop concepts that apply to a wide range of intelligences, from bacteria to superintelligences

World-centric framing and robustness to scale

- Normative agency starts from the agent's subjective viewpoint (beliefs, preferences, choice); embedded agency asks how that picture maps onto a world where the agent is just another physical subsystem
- Descriptive foundations starts from the other direction: from properties of the *world* (modularity, selection pressures, resource constraints), it asks what processes reliably steer the world into narrow targets
- **Scaling up:** our concepts must not break when the agent becomes vastly more capable – a safety property that fails under extreme optimisation pressure is useless
- **Scaling down:** the same framework should also degrade gracefully when applied to simpler systems



Descriptive Agent Foundations

Physical constraints shape agent structure

- Descriptive foundations emphasize selection pressures, constraints from the physical world, and computational boundedness – these determine what kinds of agents actually arise
- For instance, a key property of the world is **modularity**: the world decomposes into subsystems that interact sparsely. This should make both planning and world-modelling easier for any agent that exploits it
- **World-modelling**: Bayesian networks exploit modularity – instead of maintaining a full joint hypothesis over every possible world, updates about variables are only propagated locally along edges in the graph
- **Planning**: General-purpose search exploits modularity – if the world is modular, an agent can pursue decoupled subgoals independently, rather than having to plan everything jointly



Optimization and Thermodynamics

Optimizers create convergent attractors

- Keep two things apart: the **system being optimized** (its configuration could end up many ways) and the **optimizer** (a separate physical thing whose presence drives that system to a narrow target). The optimizer is not the attractor – it is its *cause*
- Operationally: an optimizer is something such that, once you **condition on its being present**, your uncertainty about where the optimized system ends up becomes *small*
- Example – a **courier** delivering a package. Left alone the package goes nowhere; the courier drives it to a fixed address from *any* starting point in the city, and **reroutes** around closures – making that address a *convergent attractor* for the package
- Two load-bearing features: a **broad range of initial conditions** all funnelled to the target, and **robustness to perturbation**. (Goals, preferences, internal representations are *not* required – this is a fact about the effect on the world)



Optimization and Thermodynamics

Even a pure predictor should attend to optimizers

- Suppose you care *only* about predicting the end state. Optimizers turn out to be, information-theoretically, the **cheapest things to know about**
- **No optimizer:** to predict where the system lands you must pin down the initial condition in detail ($\approx H(X)$ bits, enormous) *and* integrate the dynamics – and the forecast *degrades* over time as chaos amplifies any error
- **With an optimizer:** two wins. (1) the initial condition no longer matters – every start is driven to the target T , so uncertainty in X does not propagate to the outcome; (2) **naming the target is the forecast** – “it ends at T ” is a handful of bits, and *robust* (it survives perturbations you never modelled)
- Courier: tracking every vehicle and traffic light and integrating forward is hopeless; learning “the courier delivers to 14 Elm Street” is a few bits and a confident prediction
- **Upshot:** the bare aim of prediction already singles out optimizers – for *every* predictor, whatever its goals. So optimization is an **objective, observer-independent** feature of the world, worth a precise “true name”



Optimization and Thermodynamics

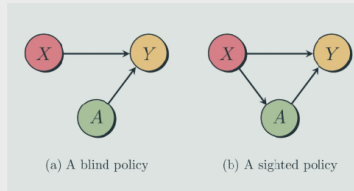
Optimization as local entropy reduction

- Powerful agents reliably steer the world towards narrow regions of target configurations – outcomes that would be extremely unlikely to arise under any random process
- This is a form of *local entropy reduction*: concentrating probability mass from a broad initial distribution onto a narrow final distribution around a target
- Irreversible transitions: funnelling many initial states into the same targets
- Existing results in stochastic thermodynamics such as fluctuation theorems (Second law of thermodynamics), cost of information erasure and information thermodynamics directly constrains the shape of optimization processes and hold far from equilibrium



Optimization and Thermodynamics

Steering costs information (Touchette & Lloyd)



- A *sighted* agent observes X and chooses action A accordingly; a *blind* agent acts independently of X . The entropy reduction a sighted agent can achieve over the blind baseline is bounded:

$$\Delta H \leq \Delta H_{\text{Blind}}^{\text{Max}} + I(X; A)$$

- The degree to which an agent can steer outcomes beyond what a blind policy achieves is paid for by the mutual information between its observations and its actions – optimization requires information



Optimization and Thermodynamics

From ensembles to individual states

- Traditional thermodynamics defines entropy in terms of probability distributions over ensembles – but where does the distribution come from?
- **Algorithmic thermodynamics** (Ehtekar & Hutter) replaces Shannon entropy H with Kolmogorov complexity K , yielding thermodynamic laws that apply to *individual physical states* rather than ensembles
- Intuitively, a state in which all particles are concentrated in one location would have low entropy, because the repeated coordinates can be printed by a short program.

Knowledge as an endogenous physical resource

- In the Gibbs-Shannon framework, a distribution μ is an *exogenous* parameter specifying what the agent knows. But an embedded agent's knowledge must be *physically encoded* in its memory
- In algorithmic thermodynamics, knowledge becomes *endogenous*: it manifests as **algorithmic mutual information** between the agent's memory and the environment
- This plays the same role as the mutual information in Touchette & Lloyd: it is the resource an embedded agent can spend to steer the world into narrow target configurations
- So an agent that *knows more* about its environment can perform *more* optimization – knowledge is the budget for steering



Optimization and Thermodynamics

Why “knowing more” formally means “optimizing more”

- Common intuition: an agent that understands its environment better can do more to it. Two bridges make this precise
- **Beliefs** → **mutual information**. If the agent's memory a encodes a belief q over environments, the true state s has a Shannon codeword of length $\approx \log \frac{1}{q(s)}$, so its description given a is short:

$$K(s | a) \leq \log \frac{1}{q(s)}, \text{ hence}$$

$$I(a; s) = K(s) - K(s | a) \geq K(s) - \log \frac{1}{q(s)}$$

- **Mutual information** → **optimization**. By (the algorithmic) Touchette & Lloyd, the agent can reduce the environment's entropy by up to $I(a; s)$
- **Together**: assigning more probability to the *true* environment $\Rightarrow$ shorter description $\Rightarrow$ more mutual information $\Rightarrow$ more optimization. *Knowing more is being able to do more* – now a theorem, with a fixed exchange rate



Conclusion

Conclusion

- We need robust concepts about agency to reason about systems far more capable than anything we have seen, and we need safety properties that are invariant under self-modification.
- **Tiling agents:** directly about making safety a reflectively stable property – ensuring that an agent's successors preserve the same guarantees, indefinitely
- **Logical uncertainty:** reasoning about mathematical and computational facts you have not yet had the compute to settle, in an environment larger than yourself. Also relevant for reasoning about your own future, more-computed beliefs – a form of self-reference. Logical induction is one concrete realisation
- **Decision theory** (treated separately): a further embedded subproblem – counterfactuals are not well-defined when you are part of the environment, which may contain copies of you or predictors modelling your decision procedure
- **Descriptive agent foundations:** starts from properties of the world (modularity, selection pressures, thermodynamic constraints) and derives what kinds of optimization processes arise – what agents embedded in the physical world would look like mechanistically
- **Connecting the two viewpoints:** reasoning about how properties of the world constrain the structure of optimization processes may tell us how an ideal agent *should* model the world and plan its actions from its subjective point of view